

CPECGM2: MINI PROJECT

Instructions

DATASET 1:

For this mini project, your task is to identify a real-world dataset that allows you to demonstrate your skills in data exploration, visualization, manipulation, and insight extraction.

Your selected dataset should be rich enough to support meaningful analysis and should enable you to:

- ✓ Explore the structure, variables, and trends within the dataset.
- ✓ Visualize key patterns, distributions, and relationships using appropriate charts or graphs.
- ✓ Manipulate the data through filtering, grouping, transforming, or cleaning techniques to prepare it for analysis.
- ✓ Extract meaningful insights—highlight interesting findings, correlations, or anomalies that provide value or provoke further questions.

Choose a dataset that aligns with your interests or a relevant real-world issue (e.g., health, education, environment, technology, or social media). You may use publicly available datasets from sources like Kaggle, data.gov.ph, UCI Machine Learning Repository, or similar platforms.

DATASET 2:

The data set `nlschools` (in the `MASS` library, use `data(nlschools, package = "MASS")` to load in the data) contains the following records for 2287 students in the Netherlands (see `?nlschools` for more information)

- `lang`: their test score on a language exam
- `IQ`: their verbal IQ
- `class`: the `tID` number for their classroom
- `GS`: the number of students in each class
- `SES`: the socio-economic status of their family
- `COMB`: was the student in a multi-grade class? (0=no, 1=yes)

Your task is to explore and describe this data set. You should use both models and visualization. You should use your judgement as to what are the most

important aspects of the data to highlight; however, the 5 questions (based on your statement of purpose) are of particular interest:

SAMPLE QUESTIONS:

1. Are there discrepancies in IQ or SES in the different classes, or when grouping by multi-grade vs non-multi-grade classes?
2. When did students perform better or worse on the language exam? Describe which variables had the most important effects.
3. Do you think there are interactions in the effects of the variables on the language exam score? Speculate as to the cause of any such effects that you think should be included.

Always think about what you are doing – blindly replicating the lectures may not result in the best analysis! Every data set is different, so you may need to use your judgement on the right thing to do. Your goal is to understand the data as best as you can, particularly about questions stated questions.

DOCUMENTATION REQUIREMENTS:

I. INTRODUCTION

This includes the brief description of the dataset to be explored. The statement of purpose and the intention of the data explorations. It should also present your main steps/procedure, and your findings.

II. ANALYSIS

This will cover the different data exploration using some statistical R commands. Create at least 5 questions to be answered in reference to the interpretation of the analyzation performed.

III. METHODS

State the different methods employed.

IV. RESULTS

Provide different data exploration in the form of plots. This should support the stated purpose of the data explorations. Make sure that all plots should have a narrative stating the insights discovered.

V. CONCLUSION

Present your conclusion based on the stated purpose and the results of your analysis and data explorations.